

What is Operation Breakdown in Garment Industry

Each garment is constructed by number of stitching jobs and non-sewing jobs. Jobs or activity involved in making a garment is known as operations. In industrial garment manufacturing, prior to making the garments, the sample is studied and list of required operations for making the approved design are made. The operations are written in a sequence of actual process flow to be followed while making the garment in the shop floor.

What is operation breakdown?

The method of preparing operations' list in a sequence is called as operation breakdown. The sheet of listed operations of a style is also known as operation breakdown.

An operation breakdown includes information like

- Sewing and non-sewing operations
- Name of the machines to use for doing the specific operations
- Estimated time to do each operation for one unit

One sample operation breakdown of Polo shirt is shown in the below image.

Operation Breakdown (Polo Shirt)

SL. No.	OPERATION	SAM	MACHINE TYPE
Cutting Room Operations			
1	Fuse Placket	0.13	Fusing M/c
2	Hem Front Placket	0.55	SNLS
3	Bottom Hem	0.64	Flat lock w/ trimmer
4	Size label to main label	0.22	SNLS w/ UBT
5	Main label to yoke	0.28	SNLS w/ UBT
Stitching Operations			
1	Attach placket to front	0.83	SNLS
	Cut placket		Manual
2	Front placket edge stitch	0.51	SNLS
3	Sew Box at placket	0.68	SNLS w/ CP
4	Yoke attach to back	0.46	DNLS
5	Shoulder overlock	0.56	5T O/L
6	Topstitch Shoulder	0.34	SNLS
7	Tack piping at collar ends		
	Attach piping to collar	0.77	SNLS
8	Turn Placket ends inside out		
	Finish Piping	1.00	SNLS
9	Cuff O/L	0.45	5T O/L
10	Cuff Topstitch	0.59	SNLS
11	Sleeve attach	0.89	5T O/L
12	Sleeve topstitch	0.60	SNLS
13	Slits Making	1.40	SNLS
14	Side seam O/L & bottom placket O/L	0.86	4T O/L
15	Tack at sleeve ends	0.30	SNLS
Finishing Operations			
20	Button Hole	0.29	B/H m/c
21	Button attaching	0.30	Buttoning M/c
22	Thread trimming	0.35	Manual
23	Garment inspection	1.45	Manual
24	Ironing the garment		Manual
25	Packing	0.43	Manual
Total SAM		12.65	

Sample operation breakdown sheet of a Polo Shirt

The garment operation bulletin is also called as operation breakdown. An operation bulletin includes more information than the operation breakdown.

Application of operation breakdown

- The operation breakdown of a garment is done to understand garment construction - like stitch class and seam types used in making the sample garment.
- To make a operation bulletin the first step is doing operation breakdown of the sample garment.
- To make the list of sewing machines and equipment for the style.
- To set set a line for new style operation breakdown is followed by line supervisors and engineers.

- While estimating thread consumption per garment for given style, operation breakdown id referred for operation list and machine types.

Operation Bulletin of a Polo Shirt

Operation bulletin, operations breakdown, machine requirement, SAM and manpower requirement are shown in this post.



Polo shirt sample

Operation Bulletin of a Polo Shirt

An operation bulletin of a polo shirt is shown in the following table. In this OB, operations list, Estimated time, machine requirement, manpower requirement and hourly production target quantity at 80% efficiency are shown.

Operation Bulletin - Polo Shirt						
Parameters			Sample garment (Polo Shirt)			
Style	Polo Shirt					
Target output	800					
Target Efficiency (Start up)	80%					
No. of Workplaces	30					
Minutes per Day	480					
Absenteeism	10%					
Operators - Sewing	24					
Total SAM	13.05					
Pieces/Op / day	33					
S.No.	Operations	Estimated Time	Machine Type	No. of operators (Calculated)	No. of machines (Calculated)	Hourly Target @80%
1	Fuse Placket	0.13	Fusing M/c	0.30	0.27	366.41
2	Hem Front Placket	0.55	SNLS	1.27	0.00	87.27
3	Bottom Hem	0.64	Flatlock	1.47	1.33	75.47
4	Size label to main label	0.22	SNLS	0.50	0.45	223.26
5	Main label to yoke	0.28	SNLS	0.64	0.58	172.66
6	Attach placket to front	0.83	SNLS	1.93	1.74	57.62
7	Cut placket	0.40	Manual	0.93	0.83	120.00
8	Front plkt edge stitch	0.51	SNLS	1.18	1.06	94.12
9	Sew Box at plkt	0.68	SNLS	1.58	1.42	70.38
10	Yoke attach to back	0.46	DNLS	1.06	0.95	105.03
11	Shoulder overlock	0.56	5T O/L	1.29	1.16	86.02
12	Topstitch Shoulder	0.34	SNLS	0.79	0.71	141.18
13	Tack piping at collar ends and Attach piping to collar	0.77	SNLS	1.79	1.61	62.18
14	Turn Placket ends inside out and Finish Piping	1.00	SNLS	2.31	2.08	48.00
15	Cuff O/L	0.45	5T O/L	1.05	0.94	106.19
16	Cuff Topstitch	0.59	SNLS	1.37	1.24	80.94
17	Sleeve attach	0.89	5T O/L	2.05	1.85	54.18
18	Sleeve topstitch	0.60	SNLS	1.38	1.24	80.40
19	Slits Making	1.40	SNLS	3.24	2.92	34.29
20	Side seam O/L & bottom plkt O/L	0.86	4T O/L	2.00	0.00	55.68
21	Tack at sleeve ends	0.30	SNLS	0.70	0.63	158.42
22	Button Hole	0.29	BH	0.67	0.60	166.09
23	Button Attaching	0.30	BS	0.70	0.63	158.94
Total SAM		13.05		30.20	24.24	
Machine	SAM	No. of machines	Actual M/c			
SNLS	8.07	18.69	19			
4T O/L	0.86	2.00	2			
DNLS	0.46	1.06	1			
Fusing M/c	0.13	0.30	0			
BS	0.30	0.70	1			
BH	0.29	0.67	1			
5T O/L	1.90	4.39	4			
Flatlock	0.64	1.47	1			
MANUAL	0.40	0.93	1			
	13.05	30.20	30			

The estimated time (SAM) shown in the above table will vary according to the product

design, method followed and machine and equipment used in actual production. So be careful while you use the above estimated time against each operation.

Operation Bulletin of a Boxer Shorts

SHARE:

[Prasanta Sarkar Dec 22, 2016](#)

Operation bulletin, operation breakdown, machine requirement and manpower requirement for making boxer shorts.

In our industrial engineering tutorial series, we are sharing operation bulletins for various products. In today's post we are sharing an operation bulletin of a boxer shorts. You can refer this operation bulletin to learn about various operations involved and machine requirement in making woven boxers shorts. If you are wondering how to make an operation bulletin read [the process of making an operation bulletin](#).



Boxer Shorts/image source shopping.indiatimes.com

Operation Bulletin

An operation bulletin of a boxer shorts is shown in the following table. In this OB, operations list, Estimated time, machine requirement, manpower requirement and hourly production target quantity at 65% efficiency are shown.

Operation Bulletin - Boxer (shorts)						
Parameters			Sample Sketch (Boxer shorts)			
Style	Boxer (back pocket)					
Target output	400					
Target Efficiency (Start up)	65%					
No. of Workplaces	19					
Minutes per Day	480					
Absenteeism	10%					
Operators - Sewing	16					
Total SAM	13.56					
Pieces/Op / day	25					
S.No.	Operation	Estimated Time	Machine type	No. of operators (calculated)	No. of machines (Calculated)	Hourly Target @65%
1	Hem Pocket opening	0.30	SNLS	0.43	0.38	130.00
2	Pocket shape press with template	0.60	IRON	0.85	0.00	65.00
3	Attach pocket to back panel	1.00	SNLS	1.42	1.28	39.00
	Front					
4	Attach right placket facing	0.62	SNLS	0.88	0.79	62.90
5	Attach facing to left placket	1.52	SNLS	2.17	1.95	25.66
	Assembly					
6	Front rise join	0.20	4THOL	0.28	0.26	195.00
7	Back rise join	0.35	4THOL	0.50	0.45	111.43
8	Top Stitch front and back rise	0.80	DNLS	1.14	1.03	48.75
9	Side seam	0.60	4THOL	0.85	0.77	65.00
10	Top Stitch side seam	0.91	DNLS	1.30	1.17	42.86
11	Make slit @ side seam bottom	1.00	SNLS	1.42	1.28	39.00
12	Finish placket and topstitch	1.20	SNLS	1.71	1.54	32.50
13	Join placket @ waistband	0.36	SNLS	0.51	0.46	108.33
14	Measure and cut elastic	0.40	MANUAL	0.57	0.00	97.50
15	Attach elastic at Waistband	0.60	4THOL	0.85	0.77	65.00
16	Waistband finish	1.00	DNCS	1.42	1.28	39.00
17	Bottom Hem	0.75	SNLS	1.07	0.96	52.00
18	Attach label (size/main)	0.50	SNLS	0.71	0.64	78.00
19	Bar tacking @slit top, pocket opening	0.45	Bartack	0.64	0.58	86.67
20	Button Hole	0.20	BH	0.28	0.26	195.00
21	Button attach	0.20	BS	0.28	0.26	195.00
	Total SAM	13.56		19.32	16.10	
Machine	SAM	No. of machines	Actual M/c			
SNLS	7.25	10.33	10			
4THOL	1.75	2.49	2			
DNLS	1.71	2.44	2			
DNCS	1.00	1.42	1			
BS	0.20	0.28	1			
BH	0.20	0.28	1			
Bartack	0.45	0.64	1			
MANUAL	0.40	0.57	1			
IRON	0.60	0.85	1			
	13.56	19.32	20			

The estimated time (SAM) shown in the above table will vary according to the product design, method followed and machine and equipment used in actual production. So be careful while you use the above estimated time against each operation.

Full form of machine name.

In the above table short name of sewing machines are written. Followings are the abbreviation of those short machine codes.

SNLS - Single needle lock stitch

4THOL - 4 thread over lock

DNLS - Double needle lock stitch

DNCS - Double needle chain stitch

BS - Button stitching

BH - Button holing

Bartack - Bar tacking machine

[HomeIndustrial EngineeringOperation bulletin](#)

Operation Bulletin of a Hoodie (Fleece Pullover)

List of sewing operations, estimated operations time and machine types required for making fleece hoodie.



Hoodie samples (Source:google image search)

Operation Bulletin of a Hoodie

An operation bulletin of a fleece hoodie is shown in the following table. In this OB, operations list, Estimated time, machine requirement, manpower requirement and hourly production target quantity at 80% efficiency are shown.

Operation Bulletin - Fleece Hoodie

Parameters			Sample garment (Fleece Hoodie)			
Style	Hoodie					
Target output	800					
Target Efficiency (Start up)	80%					
No. of Workplaces	45					
Minutes per Day	480					
Absenteeism	10%					
Operators - Sewing	41					
Total SAM	19.57					
Pieces/Op / day	20					
S.No.	Operations	Estimated Time	Machine Type	No. of operators (Calculated)	No. of machines (Calculated)	Hourly Target @80%
Preparatory						
1	Eye let attach @ hoody (2 no.)	0.28	Eyelet M/c	0.65	0.58	171
2	Make bottom and cuff ribs	0.60	SNLS	1.39	1.25	80
Hood						
1	Join inner and outer hood panels (4 seams)	1.25	SNLS	2.89	2.60	38
2	Flat lock @ hoody panels joint	1.63	Flat Lock	3.77	3.40	29
3	Join inner and outer hoody @hood opening	0.68	SNLS	1.57	1.42	71
4	Top Stitch @ hoody opening	0.70	SNLS	1.62	1.46	69
5	Stay Stitch hoody @ neck line	0.61	SNLS	1.41	1.27	79
Front						
1	Hem Kangaroo pocket opening	0.56	Flat Lock	1.30	1.17	86
2	Marking for pocket position	0.15	Manual	0.35	0.31	320
3	Attach pocket to Front	0.51	SNLS	1.18	1.06	94
4	Top Stitch pocket (3 side)	0.47	Flat Lock	1.09	0.98	102
5	Stay Stitch @ bottom	0.24	SNLS	0.56	0.50	200
Assembly						
3	Shoulder Join	0.36	4TOL	0.83	0.75	133
4	Attach Sleeve	0.86	4TOL	1.99	1.79	56
5	Flat lock @ armhole	0.90	Flat Lock	2.08	1.88	53
6	Side seam and under arm	1.09	4TOL	2.52	2.27	44
7	Attach bottom self rib to Bottom	0.92	SNLS	2.13	1.92	52
8	Attach cuff	0.66	SNLS	1.53	1.38	73
9	Over edge stitch cuff and bottom rib	0.99	4TOL	2.29	2.06	48
10	Flat lock Bottom and cuff	1.36	Flat Lock	3.15	2.83	35
11	Attach hood to neck	0.61	SNLS	1.41	1.27	79
12	Over edge stitch neck	0.36	4TOL	0.83	0.75	133
13	Attach Neck Tape	0.49	SNCS	1.13	1.02	98
14	Finish neck tape	0.50	SNLS	1.16	1.04	96
15	Top Stitch Neck	0.48	SNLS	1.11	1.00	100
16	Tack draw cord @ hood	0.24	SNLS	0.56	0.50	200
17	Tack in bottom ribX 1	0.20	Bartack M/c	0.46	0.42	240
18	Bar tack @ neck (2 Nos)	0.32	Bartack M/c	0.74	0.67	150
19	Bar tack hood X 2	0.32	Bartack M/c	0.74	0.67	150
20	Tack Pocket opening X 4	0.64	Bartack M/c	1.48	1.33	75
22	Button Hole	0.29	BH	0.67	0.60	166
23	Button Attaching	0.30	BS	0.70	0.63	159
Total SAM						
		19.57		45.30	40.77	
Machine	SAM	No. of machines	Actual M/c			
SNLS	8.00	18.52	19			
4TOL	3.66	8.47	8			
DNLS	0.00	0.00	0			
SNCS	0.49	1.13	1			
BS	0.30	0.70	1			
BH	0.29	0.67	1			
Bartack M/c	1.48	3.43	3			
Flat Lock	4.92	11.39	11			
Eyelet M/c	0.28	0.65	1			
Manual	0.15	0.35	0			
	19.57	45.30	45			

The estimated time (SAM) shown in the above table will vary according to the product design, method followed and machine and equipment used in actual production. So be careful while you use the above estimated time against each operation. The number of machines and manpower requirement depend on the product SAM and daily [production target](#).

Full form of machine name.

In the above table short name of sewing machines are written. Followings are the abbreviation of those short machine codes.

SNLS - Single needle lock stitch

4THOL - 4 thread over lock

SNCS - Single needle chain stitch

DNLS - Double needle lock stitch

BS - Button stitching

BH - Button holing

Flatlock - Flat lock machine (cover stitch)

Operation Bulletin of a Formal Trouser

Trouser operation breakdown, SAM of a formal trouser, an OB of a formal trouser.



Operation Bulletin of a Formal Trouser

An operation bulletin of a formal trouser is shown in the following table. In this OB, operations list, estimated time, machine requirement, manpower requirement and hourly production target quantity at 80% efficiency are shown.

Operation Bulletin - Formal Trouser						
Parameters		Sample garment (Trouser)				
Style	Formal Trouser					
Target output	800					
Target Efficiency (Start up)	80%					
No. of Workplaces	83					
Minutes per Day	480					
Absenteeism	10%					
Operators - Sewing	73					
Total SAM	35.70					
Pieces/Op / day	11					
S.No.	Operations	Estimated Time (SAM)	Machine type	No. of operators (calculated)	No. of machines (Calculated)	Hourly Target @80%
1	Front panel serging	0.60	3THOL	1.39	1.25	80.00
2	Parts serging	0.30	3THOL	0.70	0.00	157.89
3	Zipper Fly serging	0.20	3THOL	0.46	0.42	240.00
4	Fly ready right	0.24	SNLS	0.54	0.49	204.26
5	Front side pkt bone sew	0.46	SNLS	1.07	0.96	103.67
6	WB iron and fly iron	0.74	IRON	1.71	1.54	64.86
7	Back pkt piping	0.51	SNLS	1.18	1.06	94.12
9	Bottom hem	0.40	3THOL	0.93	0.83	120.00
10	WB fit label attach	0.41	SNLS	0.95	0.85	117.07
11	Front loading	0.58	MANUAL	1.34	1.21	82.62
12	Fly iron	0.35	IRON	0.81	0.73	137.14
15	Front pkt facing attach	0.39	SNLS	0.90	0.81	123.08
16	WB wash care label attach	0.32	SNLS	0.74	0.67	150.00
18	Front pkt bag top stitch	0.63	SNLS	1.47	1.32	75.71
19	Front pkt edge clean	0.51	SNCS	1.18	1.06	94.12
20	Right fly attach to lining	0.49	SNLS	1.13	1.02	97.96
21	Left fly attach to kinari	0.53	SNLS	1.23	1.10	90.57
22	Back panel OL	0.76	3THOL	1.76	1.58	63.16
23	Zipper attach to fly	0.63	SNLS	1.46	1.32	75.95
24	WB extension preparation	0.46	SNLS	1.06	0.00	104.35
25	WB attaching to lining	0.39	SNLS	0.90	0.81	123.08
26	Front Side Pkt saddle	0.92	Saddle Stitch	2.14	1.92	52.00
27	WB top stitch	0.42	SNLS	0.97	0.88	114.29
28	Front bar tacking	0.48	Bartack	1.11	1.00	100.00
29	Front extn. Line sewing	0.65	SNLS	1.50	1.35	73.85
30	Side pkt att+bone L	0.75	SNLS	1.74	1.56	64.00
31	Loading back panel	0.48	MANUAL	1.11	1.00	99.79
32	Side pkt att + bone R	0.75	SNLS	1.74	1.56	64.00
33	Back dart making	0.42	MANUAL	0.97	0.88	114.29
34	Dart iron	0.23	IRON	0.54	0.48	206.90
35	Back pkt welting	0.56	WELT	1.30	1.17	85.26
36	Back pkt welt iron	0.70	IRON	1.62	1.46	68.57
37	Back pkt facing att	0.30	SNLS	0.69	0.63	160.00
38	Back pkt bone sew	0.42	SNLS	0.97	0.88	114.29
39	Back pkt finish	0.79	SNLS	1.82	1.64	61.15
40	Back pkt button hole	0.44	BH	1.02	0.92	109.09
41	Back pkt bar tack	0.46	Bartack	1.07	0.96	104.12
42	Back pkt bag close	0.79	SNLS	1.83	1.65	60.76
43	Back pkt button stitch	0.39	BS	0.91	0.82	121.83
44	Brand label attach	0.30	SNLS	0.69	0.63	160.00
45	Panel pairing	0.50	MANUAL	1.16	1.04	96.00
46	Side seam SNCS	0.72	SNCS	1.67	1.50	66.67
47	Inseam SNCS	0.67	SNCS	1.56	1.40	71.43
48	Crotch sew	0.70	SNLS	1.62	1.46	68.47
49	Right fly attach to zipper	0.66	SNLS	1.53	1.37	72.84
51	Loop saddle stitch	0.30	Saddle Stitch	0.69	0.63	160.00
52	Loop attach	0.85	SNLS	1.97	1.77	56.47
53	WB attach	0.80	SNLS	1.85	1.67	60.00
54	Hook and bar attach	0.56	Hook-Bar	1.29	1.16	86.33
55	WB pick ready	0.89	SNLS	2.06	1.85	53.93
56	Fly iron	0.21	IRON	0.49	0.44	228.57
57	Fly kinari	0.63	SNLS	1.46	1.31	76.19
58	Fly top stitch	0.62	SNLS	1.44	1.29	77.42
59	Seat seam turn	0.69	MANUAL	1.59	1.44	69.67
60	WB iron	0.63	IRON	1.46	1.31	76.19
61	WB blind hem	0.40	Blind hem	0.93	0.84	119.70
62	WB Extension iron	0.39	IRON	0.90	0.81	123.08
63	Loop down bartack	0.54	Bartack	1.25	1.13	88.89
64	Loop top bar tack	0.63	Bartack	1.47	1.32	75.71
65	Loop top & down bartack	0.98	Bartack	2.27	2.04	48.98
66	Spot tacking	0.40	Bartack	0.93	0.83	120.00
67	Fly bartack	0.50	Bartack	1.16	1.04	96.00
68	WB button holing	0.43	BH	1.00	0.90	110.60
69	Bottom hem	0.68	3THOL	1.57	1.42	70.59
70	WB Button Sew	0.40	BS	0.93	0.83	120.00
8	End of line inspection	0.75	MANUAL	1.74	1.56	64.00
Total SAM		35.70		82.65	72.79	
Machine	SAM (Machine wise)	No. of machines (Calculated)	Actual M/c number			
SNLS	15.78	36.53	37			
SNCS	1.39	3.22	3			
SNCS	0.51	1.18	1			
3THOL	2.94	6.81	7			
BS	0.79	1.84	2			
BH	0.87	2.02	2			
Hook-bar	0.56	1.29	1			
WELT	0.56	1.30	1			
Saddle Stitch	1.22	2.83	3			
Blind hem	0.40	0.93	1			
Bartack	4.00	9.25	9			
IRON	3.25	7.53	8			
MANUAL	3.42	7.92	8			
Total	35.70	82.65	83			



Sample operation bulletin - Formal trouser

The [standard time \(SAM\)](#) shown in the above OB, may vary depending on the product design, methods followed, and machine and equipment used in actual production. So be careful while you

measure SAM against each operation. You are suggested to calculate operations SAM by your own if you know how to calculate the operation SAM.

Read other Operations Bulletins

Full form of the machine names

In the above table, the short name of sewing machines is written. Followings are the abbreviation of those short machine codes.

SNLS - Single needle lock stitch

3THOL - Overlock (3 thread)

SNCS - Single needle chain stitch

BS - Button sewer

BH - Button hole machine

Hook-bar - Hook and bar attaching machine

SNEC - Single needle edge cutter

WELT - Auto pocket welting machine

Saddle stitch - Saddle stitch machine

Blind hem - Blind hemming machine

Bartack - Bartack machine

Operation Bulletin for Panties (Inner Elastic Panty)

Operation breakdown of panties with list of sewing machines.

Panties are inner wear products for Ladies. Panties are mostly made of fine knits fabrics. Panties are available in various design and fit.

Product components: This panty is made of 4 components - Front panel (1), Back panel (1) and crotch panels (2).

Trims: Waist band elastic, sewing thread, labels, and elastic for leg opening.



Pic: Inner elastic panty (image source: cilory.com)

Sample Operation Bulletin of Panties (Inner Elastic)

S.No.	Operation	SAM	Machine type
1	Attach crotch panel with front panel	0.20	3 Thread Over lock
2	Attach crotch panel with back panel	0.20	Single Needle Lock Stitch
3	Attach side seam	0.30	3 Thread Over lock
4	Measure and cut elastic	0.40	Manual
5	Join elastic ends	0.20	Single Needle Lock Stitch

6	Attach elastic at Waistband	0.60	3 Thread Over lock
7	Waistband finish Flat lock	0.69	2 Needle Flat Lock
8	Bottom hem with elastic tape	0.80	2 Needle Flat Lock
Total SAM		3.39	

Note: List of operations and machine type to be used may vary based on product design and its construction. Example: You can use Zig-Zag sewing machine for attaching elastic ends instead of Lock stitch machine. In some styles, bar tack stitch is used to secure stitch.

Operation Breakdown: Bra

OP #	SEWING PROCESS	SAM	MACHINE
1	Join Cups	0.57	SNLS
2	Attach Tape to Cups	0.73	DNLS
3	attach Elastic to side cups	0.44	3SZZ
4	Set support Panels	0.39	SNLS
5	Serge support panels	0.62	3TOL
6	Join support panels, cup and front	1.52	SNLS
7	Attach coverstitch tape to front	0.61	DNLS
8	Join backs to cups	0.55	SNLS
9	Attach coverstitch tape to sides	0.47	DNLS
10	Attach lace and tape to cups	0.76	DNLS
11	Attach elastic to band	0.90	ZZ
12	Attach upper elastic & straps	0.91	3SZZ

13	Cover support panels edges	0.54	3TOL
14	Attach hooks and eyes to straps	1.39	SNLS
15	Set eye to back	0.36	ZZ
16	Set hook to back	0.50	SNLS
17	Attach trimming to front	0.25	SNLS
18	Trim thread and inspect	1.00	MAN
	Total SAM	12.51	

Full name of the sewing machine

SNLS: Single Needle Lock Stitch machine

DNLS: Double Needle Lock stitch machine

3TOL: 3 thread over lock / over edge sewing machine

ZZ: Zig-zag sewing machine (3S for 3 stitch)

MAN: Manual work

Operation Breakdown and SMVs of a Basic Jeans

Operation breakdown of a jeans pant with SMV.

Operation breakdown of a 5 Pocket Basic Jeans is listed in this article. SMVs mentioned here against each operation are just for your reference. SMVs may vary according to machine types, workstation layout and equipment used.

Some of operations are done by automatic machines, like auto pocket hemming (APH), auto pocket attaching (APS) etc. Machines that are used for the operations have been mentioned in third column (M/c type) of the following table.

This Operation Breakdown has been taken from a Jeans manufacturing company in Vietnam.



5-Pocket Basic Jeans

Table 1: Operation Breakdown Chart

Sl. no.	Operations	M/C Type	SMV
Back			
1	pocket hemming	APH	0.07
2	pocket o/l	3T O/L	0.3
3	pocket creasing	ADPC	0.4
4	pocket attaching	APS	0.74
5	second stitch @ back pocket	SNLS	0.65
6	back yoke attach	FOA	0.41
7	back rise join	FOA	0.24
8	size label attach	SNLS	0.2
Front Section			
9	coin pocket hemming	APH	0.015
10	coin pocket attach @ jet piece (right)	SNLS	0.2

11	pocket bag attach with jet piece	5T F/L	0.42
12	o/l @ pocket bag	3T O/L	0.68
13	top stitch @ pocket bag	SNLS	0.4
14	o/l @ zipper fly	4T O/L	0.3
15	o/l front panels @ crotch	3T O/L	0.26
16	zipper attach @ left fly	DNLS	0.32
17	left fly attach @ front panel (inseam & top stitch)	SNLS	0.34
18	J-stitch @ left fly	APJ	0.58
19	front pocket mouth hemming	DNFPH	0.3
20	pocket bag stitch to front panel @side & top (wt w/c label@lt.)	SNLS	0.46
21	right fly attach with zipper & crotch join	SNLS	0.25
Assembly			
22	loop preparation	3T F/L	0.15
23	main label attach @ waist band, & w/b joining in chain	SNLS	0.3
24	top stitch @ inseam	FOA	0.55
25	side seam join (attach front & back panels)	5T O/L	0.65
26	top stitch @ side seam	SNLS	0.38
27	waist band stitch	WBAM	0.38
28	waist band corners finish	SNLS	0.72
29	loop attach	ABLA	0.65
30	bottom hemming	BHM	0.33
	Total		11.65

Full form of sewing machine name

APH - Automatic pocket hemming machine
ADPc - Automatic pocket creasing machine
APc - Auto pocket stitching machine
SNLS - Single needle lock stitch
FOA - Feed of the arm
3TO/L - 3 thread overlock (overedge)
5TF/L - 5 thread flat lock (cover stitch)
4TO/L - 4 thread over lock
5TO/L - 5 thread over lock
DNLS - Double needle lock stitch
3TF/L - 3 thread flat lock (cover stitch)
APJ - Auto J-stitch making
DNFPH - Double needle front pocket hemming
WBAM - Waistband attach machine
ABLA - Auto loop attach
BHM - Bottom hemming machine

(Note: Some of the machine names are short form of machine used for the specific operation used by a company. Other companies may name it different way)

Operation Breakdown and SMV of a Trouser

Trouser operation breakdown

The operation breakdown of a trouser with two front pocket and two back pocket has been given in the following table. SMVs mentioned here against each operation are just for your reference. It may vary according to machine types, workstation layout and equipment used. Operations needed to make a trouser have been listed section wise. Machines that are used for the operations have been also mentioned in the table.



Trousers /Image from pinterest.com

Table 1: Operation Breakdown Chart

Section	Sequence No.	Operation Description	Mc type	Smv
Preparatory	1	Serge front panel with knee lining	Auto serge	0.22
	2	Serge back panel	Auto serge	0.38
	3	Serge right fly	3ol	0.08
	4	Serge back rise	3ol	0.32
	5	Serge front rise	3ol	0.27
	6	Make back dart 2x2	Auto dart	0.22
	7	Trim w/b	Snec	0.25
	8	Attach gripper	Gripper mc	0.45
	9	Make fly-box	Snec	0.20
	10	Fly binding	Snls	0.14
	11	Attach front pkt facing	Auto facing	0.38
	12	Attach back pkt facing	Auto facing	0.27
	13	Cut w/b	Indirect	0.12
	14	Waistband checking	Chk	
Front	16	Tack pkt bag	Snls	0.55
	17	Close pkt bag with knitted tape	5ol	0.47
	19	Loading front & small parts	Indirect	0.40
	20	Attach left fly & t/s with inside tacking lining	Snls	0.42
	21	cut & attach pocket bag @ front with lining	Snnf	0.65
	22	Press pocket & fly	Ironer	0.46
	23	Top stitch @ pkt mouth	Snls	0.54
	24	Tack pkt side & waist	Snls	0.48
	25	Attach fly box & zipper with e/s stitch @ zipper.	Snls	0.62
	26	Front checking	Chk	
Back	27	Fuse back dart	Ironer	0.46
	28	Make back bone	Apw	0.53
	29	Back loading	Indirect	0.35
	30	Press back pkt bone	Ironer	0.52
	31	Attach main label @ back pkt x1	Snls	0.44
	32	Tack back pkt corner with close facing	Snls	0.77
	33	Make eye hole	Bh	0.33
	34	Tack back pkt to pkt mouth	Snls	0.55
	35	Close pkt bag with knitted tape	5ol	0.56
	36	Back checking	Chk	
Assembly	37	Pairing	Indirect	0.45
	38	Join side seam & in seam with knee lining	Sncs	1.35
	39	Check side seam	Chk	
	40	Safety stitch	Snls	0.39
	41	Press side & in seam	Ironer	0.59
	42	Assembly loading & belt loading	Indirect	0.42
	43	Belt marking	Indirect	0.40
	44	Make round extension	Snls	0.52
	45	Crotch joint	Snls	0.46
	46	Attach & close zipper	Snls	0.42
	47	Loop marking	Indirect	0.48
	48	Loop attach	Snec	0.76
	49	Attach w/b	Snuf	1.05
	50	Make extension	Snnf	0.53

Table 2: Abbreviation of machine name

Machine	Description	Machine	Description	Machine	Description
SNLS	1 Needle Lock stitch	DNLS	2 Needle Lock stitch	APW	Auto Pocket welting
SNCS	1 Needle Chain stitch	DNCS	2 Needle Chain Stitch	BT	Bar tack machine
SNEC	1 Needle Edge cutter	3OL	3 thread Over lock	BH	Button Holing
TNDM NDL	Tandem needle	4OL	4 thread Over lock	BA	Button Attach
BS	Blind stitch	5OL	5 thread Over lock	CHK	Checking